Chapter 10: Multiple Integrals — Complete Solutions

1. $I = \iint\limits_D \sqrt{1-x^2-y^2}\,\mathrm{d}\sigma$, where D is the region enclosed by the curve $y = \sqrt{1-x^2}$ and the line $y = 0$

Solution:

Step 1: Determine the region of integration

Squaring both sides of the curve equation $y = \sqrt{1-x^2}$ gives $x^2 + y^2 = 1$ (with $y \geq 0$).

Hence the region D is the upper half of the unit disk: $x^2 + y^2 \leq 1, y \geq 0$.

Step 2: Convert to polar coordinates

Let $x = r\cos\theta, y = r\sin\theta$; then the area element is $\mathrm{d}\sigma = r\,\mathrm{d}r\,\mathrm{d}\theta$.

The range of integration in polar coordinates is: $0 \leq \theta \leq \pi, \quad 0 \leq r \leq 1$.

Step 3: Evaluate the integral

$$I = \int_0^{\pi} \mathrm{d}\theta \int_0^1 \sqrt{1-r^2} \cdot r\,\mathrm{d}r$$

Since the integrand depends only on r, the integral separates:

$$I = \left(\int_0^{\pi} 1\,\mathrm{d}\theta\right) \cdot \left(\int_0^1 (1-r^2)^{\frac{1}{2}} r\,\mathrm{d}r\right)$$

$$I = [\theta]_0^{\pi} \cdot \left[-\frac{1}{3}(1-r^2)^{\frac{3}{2}}\right]_0^1$$

$$I = \pi \cdot \left(0 - \left(-\frac{1}{3}\right)\right) = \frac{\pi}{3}$$

2. Sketch each region of integration and change the order of integration of each iterated integral

(1) $\int_1^2 \mathrm{d}x \int_{2-x}^{\sqrt{2x-x^2}} f(x,y)\mathrm{d}y$

Solution:

Step 1: Recover the boundary of the region of integration D

From the original limits of integration, the region D satisfies: $1 \leq x \leq 2$, and $2 - x \leq y \leq \sqrt{2x-x^2}$.

- Lower boundary: $y = 2 - x \implies x = 2 - y$ (a straight line through the points $(1,1)$ and $(2,0)$)
- Upper boundary: $y = \sqrt{2x-x^2} \implies y^2 = 2x - x^2 \implies (x-1)^2 + y^2 = 1$ (the part, lying in the first quadrant, of the circle centered at $(1,0)$ with radius 1). Solving the equation for x: $x = 1 + \sqrt{1-y^2}$ (taking the right half of the circle).

Step 2: Change the order of integration (x first, then y)

From the sketch, the two boundaries intersect at $(1,1)$ and $(2,0)$.

Projecting the region onto the y-axis, the range of y is $0 \leq y \leq 1$.

For fixed y, the left boundary (entry point) is the line $x = 2 - y$, and the right boundary (exit point) is the circular arc $x = 1 + \sqrt{1 - y^2}$.

Hence the integral with the order changed is:

$$\int_0^1 \mathrm{d}y \int_{2-y}^{1+\sqrt{1-y^2}} f(x, y)\mathrm{d}x$$

(2) $\int_1^2 \mathrm{d}x \int_{\frac{1}{x}}^1 f(x, y)\mathrm{d}y$

Solution:

Step 1: Recover the region of integration D

The region D satisfies: $1 \leq x \leq 2$, and $\frac{1}{x} \leq y \leq 1$.

The boundary curves are the hyperbola $y = \frac{1}{x}$, the horizontal line $y = 1$, and the vertical lines $x = 1$ and $x = 2$.

Step 2: Change the order of integration (x first, then y)

According to the intersection points, the hyperbola meets $x = 2$ at $(2, \frac{1}{2})$ and $x = 1$ at $(1, 1)$.

Projecting the region onto the y-axis, the range of y is $\frac{1}{2} \leq y \leq 1$.

For fixed y, crossing the region from left to right, the left boundary is the hyperbola $x = \frac{1}{y}$ and the right boundary is the line $x = 2$.

Hence the integral with the order changed is:

$$\int_{\frac{1}{2}}^1 \mathrm{d}y \int_{\frac{1}{y}}^2 f(x, y)\mathrm{d}x$$

3. Compute the following double integrals

(1) $\iint\limits_D (3x - 2y)\mathrm{d}\sigma$**, where D is the closed region enclosed by the two coordinate axes and the line** $x + y = 2$

Solution:

The region D is an isosceles right triangle with vertices $(0, 0), (2, 0), (0, 2)$.

Convert to an iterated integral with y first and then x:

$$\iint\limits_D (3x - 2y)\mathrm{d}\sigma = \int_0^2 \mathrm{d}x \int_0^{2-x} (3x - 2y)\mathrm{d}y$$

First compute the inner integral:

$$\int_0^{2-x} (3x - 2y)\mathrm{d}y = \left[3xy - y^2\right]_0^{2-x} = 3x(2 - x) - (2 - x)^2$$

Expanding gives:

$$= 6x - 3x^2 - (4 - 4x + x^2) = -4x^2 + 10x - 4$$

Then compute the outer integral:

$$\int_0^2(-4x^2+10x-4)\mathrm{d}x=\left[-\frac{4}{3}x^3+5x^2-4x\right]_0^2$$
$$=-\frac{32}{3}+20-8=12-\frac{32}{3}=\frac{4}{3}$$

(2) $\iint\limits_D xy^2\mathrm{d}\sigma$, where D is the closed right half-region enclosed by the circle $x^2+y^2=4$ and the y-axis

Solution:

Since the region is related to a circle, polar coordinates are the most convenient.

The region D is the right half-disk: $0\le r\le 2,\quad -\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$.

The integrand becomes: $xy^2=(r\cos\theta)(r\sin\theta)^2=r^3\cos\theta\sin^2\theta$.

Substituting into the double-integral formula in polar coordinates:

$$\iint\limits_D xy^2\mathrm{d}\sigma=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^2(r^3\cos\theta\sin^2\theta)\cdot r\,\mathrm{d}r$$

Separating the variables and integrating:

$$=\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos\theta\sin^2\theta\,\mathrm{d}\theta\right)\cdot\left(\int_0^2 r^4\,\mathrm{d}r\right)$$

Computing each separately:

$$\int_0^2 r^4\,\mathrm{d}r=\left[\frac{1}{5}r^5\right]_0^2=\frac{32}{5}$$
$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2\theta\,\mathrm{d}(\sin\theta)=\left[\frac{1}{3}\sin^3\theta\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}=\frac{1}{3}(1-(-1))=\frac{2}{3}$$

Multiplying the two:

$$\iint\limits_D xy^2\mathrm{d}\sigma=\frac{2}{3}\cdot\frac{32}{5}=\frac{64}{15}$$

4. Compute the volume of the solid enclosed by the coordinate planes, the planes $x=4$, $y=4$, and the paraboloid $z=x^2+y^2+1$

Solution:

Since the base of the solid lies in the coordinate plane ($z=0$), the top boundary is the paraboloid $z=x^2+y^2+1$, and the region of integration is the square region $D=\{(x,y)\mid 0\le x\le 4, 0\le y\le 4\}$.

The volume V can be expressed as the double integral of the curved-top cylindrical solid:

$$V=\iint\limits_D(x^2+y^2+1)\mathrm{d}\sigma=\int_0^4\mathrm{d}x\int_0^4(x^2+y^2+1)\mathrm{d}y$$

Compute the inner integral:

$$\int_0^4(x^2+y^2+1)\mathrm{d}y=\left[x^2y+\frac{1}{3}y^3+y\right]_0^4=4x^2+\frac{64}{3}+4=4x^2+\frac{76}{3}$$

Compute the outer integral:

$$V = \int_0^4 \left(4x^2 + \frac{76}{3}\right) \mathrm{d}x = \left[\frac{4}{3}x^3 + \frac{76}{3}x\right]_0^4$$

$$V = \frac{4}{3}(64) + \frac{76}{3}(4) = \frac{256}{3} + \frac{304}{3} = \frac{560}{3}$$

5. Use polar coordinates to compute the following double integrals

(1) $\iint\limits_D e^{-x^2-y^2}\,\mathrm{d}\sigma$, where D is the part of the unit disk in the first quadrant

Solution:

In polar coordinates the region D is given by: $0 \le \theta \le \frac{\pi}{2}, \quad 0 \le r \le 1$.

The integrand becomes $e^{-x^2-y^2} = e^{-r^2}$.

$$\iint\limits_D e^{-x^2-y^2}\,\mathrm{d}\sigma = \int_0^{\frac{\pi}{2}} \mathrm{d}\theta \int_0^1 e^{-r^2} r\,\mathrm{d}r$$

The outer integral is immediate: $\int_0^{\frac{\pi}{2}} \mathrm{d}\theta = \frac{\pi}{2}$.

For the inner integral, recognize the differential:

$$\int_0^1 e^{-r^2} r\,\mathrm{d}r = -\frac{1}{2}\int_0^1 e^{-r^2}\mathrm{d}(-r^2) = \left[-\frac{1}{2}e^{-r^2}\right]_0^1 = -\frac{1}{2}(e^{-1} - 1) = \frac{1}{2}\left(1 - \frac{1}{e}\right)$$

Hence the original integral is:

$$\frac{\pi}{2} \cdot \frac{1}{2}\left(1 - \frac{1}{e}\right) = \frac{\pi}{4}\left(1 - \frac{1}{e}\right)$$

(2) $\iint\limits_D (4 - x - y)\,\mathrm{d}\sigma$, where D is the disk $x^2 + y^2 \le 2y$

Solution:

> 💡 **Exam tip:** The problem explicitly asks to "compute using polar coordinates", so the polar-coordinate steps must be written out. However, after finishing, it is strongly recommended to verify using **odd–even symmetry plus the centroid formula**: the region is a disk centered at $(0, 1)$ with area π; by symmetry $\iint x\,\mathrm{d}\sigma = 0$, and $\iint y\,\mathrm{d}\sigma = \bar{y} \cdot A = 1 \cdot \pi = \pi$, so the total integral is $4\pi - 0 - \pi = 3\pi$.

Formal polar-coordinate solution:

Convert the boundary $x^2 + y^2 = 2y$ to polar coordinates: $r^2 = 2r\sin\theta \implies r = 2\sin\theta$.

Since $r \ge 0$, we have $\sin\theta \ge 0$, so the polar-angle range of the region is: $0 \le \theta \le \pi$.

The integrand becomes $4 - r\cos\theta - r\sin\theta$.

$$I = \int_0^{\pi} \mathrm{d}\theta \int_0^{2\sin\theta} (4 - r\cos\theta - r\sin\theta) r\,\mathrm{d}r$$

The inner integral:

$$\int_0^{2\sin\theta} (4r - r^2\cos\theta - r^2\sin\theta)\,\mathrm{d}r = \left[2r^2 - \frac{1}{3}r^3\cos\theta - \frac{1}{3}r^3\sin\theta\right]_0^{2\sin\theta}$$

$$= 8\sin^2\theta - \frac{8}{3}\sin^3\theta\cos\theta - \frac{8}{3}\sin^4\theta$$

Substituting into the outer integral and computing term by term:

1. $\int_0^{\pi} 8\sin^2\theta\,\mathrm{d}\theta = 8 \times 2\int_0^{\frac{\pi}{2}} \sin^2\theta\,\mathrm{d}\theta = 16 \times \frac{1}{2}\frac{\pi}{2} = 4\pi$ (using Wallis' formula)
2. $\int_0^{\pi} \frac{8}{3}\sin^3\theta\cos\theta\,\mathrm{d}\theta = \frac{8}{3}\left[\frac{1}{4}\sin^4\theta\right]_0^{\pi} = 0$
3. $\int_0^{\pi} \frac{8}{3}\sin^4\theta\,\mathrm{d}\theta = \frac{8}{3} \times 2\int_0^{\frac{\pi}{2}} \sin^4\theta\,\mathrm{d}\theta = \frac{16}{3} \times \frac{3}{4}\frac{1}{2}\frac{\pi}{2} = \pi$

Combining the results:

$$I = 4\pi - 0 - \pi = 3\pi$$

6. Find the area of the part of the hemisphere $z = \sqrt{2 - x^2 - y^2}$ cut out by the cone $z = \sqrt{x^2 + y^2}$ that lies inside the cone

Solution:

> ⚠️ **Common pitfall:** What is asked for is the "area", not the "volume"; the first-kind surface integral formula $A = \iint\limits_D \sqrt{1 + z_x^2 + z_y^2}\,\mathrm{d}\sigma$ must be used.

Step 1: Find the projected region of integration D

Combine the equations of the two surfaces to find their curve of intersection:

$\sqrt{2 - x^2 - y^2} = \sqrt{x^2 + y^2} \implies 2 - (x^2 + y^2) = x^2 + y^2 \implies x^2 + y^2 = 1.$

Hence the region D projected onto the xOy plane is the unit disk: $x^2 + y^2 \leq 1$.

Step 2: Compute the area element dS of the surface

Compute the partial derivatives of the hemisphere $z = \sqrt{2 - x^2 - y^2}$:

$z_x = \frac{-x}{\sqrt{2-x^2-y^2}}, \quad z_y = \frac{-y}{\sqrt{2-x^2-y^2}}$

$1 + z_x^2 + z_y^2 = 1 + \frac{x^2}{2-x^2-y^2} + \frac{y^2}{2-x^2-y^2} = \frac{2-x^2-y^2+x^2+y^2}{2-x^2-y^2} = \frac{2}{2-x^2-y^2}$

The area element is $dS = \frac{\sqrt{2}}{\sqrt{2-x^2-y^2}}\mathrm{d}x\mathrm{d}y$.

Step 3: Evaluate the integral

Convert to polar coordinates:

$$A = \iint\limits_D \frac{\sqrt{2}}{\sqrt{2 - x^2 - y^2}}\mathrm{d}\sigma = \int_0^{2\pi}\mathrm{d}\theta\int_0^1 \frac{\sqrt{2}}{\sqrt{2 - r^2}} r\,\mathrm{d}r$$

The outer integral equals 2π. For the inner integral, recognize the differential:

$$\int_0^1 \sqrt{2}(2 - r^2)^{-\frac{1}{2}} r\,\mathrm{d}r = -\frac{\sqrt{2}}{2}\int_0^1 (2 - r^2)^{-\frac{1}{2}}\mathrm{d}(2 - r^2) = -\frac{\sqrt{2}}{2}\left[2(2 - r^2)^{\frac{1}{2}}\right]_0^1$$

$$= -\sqrt{2}\left(\sqrt{1} - \sqrt{2}\right) = -\sqrt{2} + 2 = 2 - \sqrt{2}$$

The total area:

$$A = 2\pi(2 - \sqrt{2})$$

7. Find the surface area of the solid enclosed by two mutually perpendicular circular cylinders $x^2+y^2=R^2$ and $x^2+z^2=R^2$ with the same base radius

Solution:

> 💡 **Exam tip (ultimate symmetry method):** This is the surface area of the famous "mou he fang gai" (Steinmetz) solid. The solid is pieced together from completely identical surface patches in the 8 octants (front/back, left/right, top/bottom), and the surfaces of the two cylinders are symmetric to each other. We only need to compute the surface area of one cylinder piece (say $x^2+z^2=R^2$) in the first octant, and then multiply by **16** (2 cylinders $\times$ 8 octants = 16 patches).

Step 1: Determine the single integration surface and the projected region

Consider a portion of the cylinder $x^2+z^2=R^2$ inside the first octant ($x\geq 0, y\geq 0, z\geq 0$).

Its equation is $z=\sqrt{R^2-x^2}$.

This portion of surface lies inside the other cylinder $x^2+y^2=R^2$, so its projected region D_1 on the xOy plane is:

$0\leq x\leq R,\quad 0\leq y\leq \sqrt{R^2-x^2}$.

Step 2: Compute the area element dS

Compute the partial derivatives of $z=\sqrt{R^2-x^2}$:

$z_x=\frac{-x}{\sqrt{R^2-x^2}},\quad z_y=0$

$1+z_x^2+z_y^2=1+\frac{x^2}{R^2-x^2}=\frac{R^2}{R^2-x^2}$

The area element is $dS=\frac{R}{\sqrt{R^2-x^2}}\mathrm{d}x\mathrm{d}y$.

Step 3: Compute the area S_1 of the single patch in the first octant

$$S_1=\iint\limits_{D_1}\frac{R}{\sqrt{R^2-x^2}}\mathrm{d}\sigma=\int_0^R\mathrm{d}x\int_0^{\sqrt{R^2-x^2}}\frac{R}{\sqrt{R^2-x^2}}\mathrm{d}y$$

The inner integral is:

$$\left[\frac{R}{\sqrt{R^2-x^2}}y\right]_0^{\sqrt{R^2-x^2}}=\frac{R}{\sqrt{R^2-x^2}}\cdot\sqrt{R^2-x^2}=R$$

The outer integral is:

$$S_1=\int_0^R R\,\mathrm{d}x=R\cdot R=R^2$$

Step 4: Obtain the total area

The solid is made up of two cylinder surfaces, each of which has surface patches of area R^2 in each of the 8 octants — 16 patches in total.

The total surface area is $S=16\times S_1=16R^2$.